

# Equivalent Expressions

Lesson 6-6

Name: \_\_\_\_\_

Date: \_\_\_\_\_

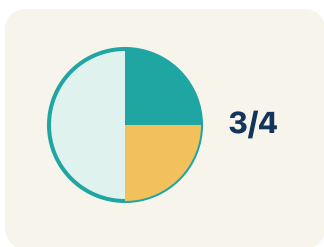
Class: \_\_\_\_\_

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## Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



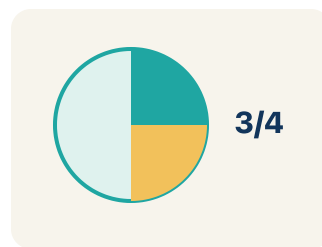
$2x + 4$  and  $2(x + 2)$ : when  $x = 3$ , both = 10; when  $x = 7$ , both = 18 — always the same

### Equivalent

Write the definition:

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$3x + 2x + 5$  simplifies to  $5x + 5$  — fewer terms, same value

### Simplify

Write the definition:

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$$3x + 2x = 5x$$

same variable → combine

$3x$  and  $7x$  are like terms (both  $x$ );  $3x$  and  $3y$  are NOT ( $x$  vs  $y$ );  $3x$  and  $3x^2$  are NOT ( $x$  vs  $x^2$ )

### Like Terms

Write the definition:

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$$2 + 3 = 5$$

$4n + 3n = 7n$  — add the coefficients ( $4 + 3 = 7$ ) and keep the variable ( $n$ )

### Combine

Write the definition:

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$$4x$$

coefficient = 4

*In  $5x + 3$ , the coefficient of  $x$  is 5 — when  $x = 2$ , the  $5x$  part equals 10*

## Coefficient

Write the definition:

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## Guided Notes

Level 2 Standard



### WHAT WE'RE LEARNING TODAY

**I can show that two expressions are equivalent by simplifying and combining like terms.**



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent

Simplify

Like Terms

Combine

Coefficient

1

\_\_\_\_\_ — Expressions that always have the same value.

2

To write an expression in its shortest equal form is to \_\_\_\_\_ it.

3

Terms with the same variable raised to the same power, like  $4x$  and  $9x$ , are

\_\_\_\_\_.

4

To add or subtract like terms into one term is to \_\_\_\_\_ them.

5

A number multiplied by a variable is a \_\_\_\_\_.



### Watch & Try — Worked Examples

*See the notes in action: watch one worked all the way through, then try the next with the same steps.*

 **I do — watch**

Follow each step as your teacher solves it.

**Problem:** Which expression is equivalent to  $4x + 3x$ ?

- A.  $7x$
- B.  $12x$
- C.  $7x^2$
- D.  $43x$

**Step 1**  $4x + 3x = 7x$ .

**Step 2** Combine the coefficients:  $4 + 3 = 7$ .

 **Answer:** A.  $7x$

 **We do — together**

Solve this one with your class using the same steps.

**Problem:** Which expression is equivalent to  $2(m + 5)$ ?

- A.  $2m + 10$
- B.  $2m + 5$
- C.  $m + 10$
- D.  $7m$

**Step 1** \_\_\_\_\_

**Step 2** \_\_\_\_\_

**Answer:** \_\_\_\_\_

 **You do — your turn**

Now try one on your own. Show every step.

**Problem:** Which property is shown:  $3(x + 7) = 3x + 21$ ?

- A. Distributive
- B. Commutative
- C. Associative
- D. Identity

Show your work:

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## Try It

Solve on your own. Check the answer key when you are done.

**1. A track stacks  $3x$  and then one more  $x$ . Which expression is equivalent to  $3x + x$ ?**

- A.  $4x$
- B.  $3x^2$
- C.  $4$
- D.  $3x$

Show your work:

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**2. Producer B flipped a formula to  $5 + 2x$ . Which expression is equivalent?**

- A.  $2x + 5$
- B.  $7x$
- C.  $10x$
- D.  $5x + 2$

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A student says  $2x + 3$  and  $5x$  are equivalent because both equal 5 when  $x = 1$ . Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of  $x$  where they give different results.**

*Sentence starter: The student is \_\_\_\_ because \_\_\_\_\_. When  $x = 1$ :  $2(1) + 3 = \underline{\hspace{1cm}}$  and  $5(1) = \underline{\hspace{1cm}}$ . But when  $x = \underline{\hspace{1cm}}$ :  $2(\underline{\hspace{1cm}}) + 3 = \underline{\hspace{1cm}}$  and  $5(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$ . To prove equivalence, expressions must be equal for \_\_\_\_ values of  $x$ .*

Show your work:

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## Reflect — Exit Ticket

**Which expression is equivalent to  $6n + 4 + 3n - 1$ ?**

- A.  $9n + 3$
- B.  $9n + 5$
- C.  $63n$
- D.  $18n + 4$

Your answer:

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## Answer Key & Teacher Guide

1. **Notes 1:** Equivalent
2. **Notes 2:** Simplify
3. **Notes 3:** Like Terms
4. **Notes 4:** Combine
5. **Notes 5:** Coefficient
6. **We Try (notes):** A.  $2m + 10 - 2(m + 5) = 2m + 10$  using the distributive property.
7. **You Try (notes):** A. Distributive — *Multiplying 3 by each term inside the parentheses is the distributive property.*
8. **Try It 1:** A.  $4x - 3x + x$  is 3 groups of  $x$  plus 1 more group of  $x = 4x$ . Add the coefficients:  $3 + 1 = 4$ . Check  $x = 3$ :  $3(3) + 3 = 12$  and  $4(3) = 12$ .
9. **Try It 2:** A.  $2x + 5$  — *Order of addition does not change the value (commutative property):  $5 + 2x = 2x + 5$ . Check  $x = 10$ :  $5 + 2(10) = 25$  and  $2(10) + 5 = 25$ .*
10. **Exit Ticket:** A.  $9n + 3 - 6n + 3n = 9n$  and  $4 - 1 = 3$ , so the simplified expression is  $9n + 3$ .